

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 39)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change

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Ozone depletion

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Tourism

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Future mineral exploitation

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0 1 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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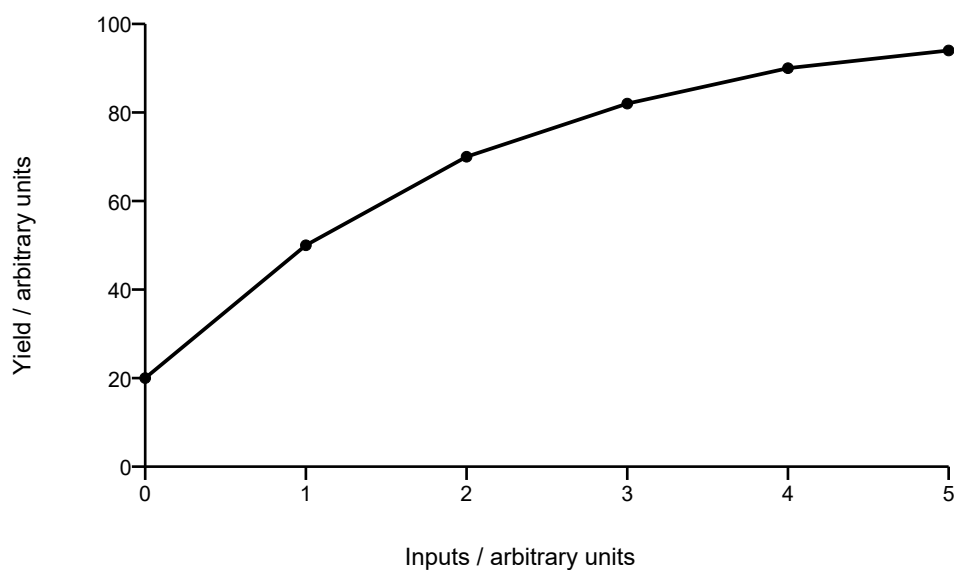
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10

0 2

Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1**0 2****1**

Describe the relationship shown in Figure 1.

[2 marks]**0 2****2**

Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]**Answer**

units

0 2**3**

Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

0 2 . 4 Suggest why inputs are nevertheless often used unevenly across the world. **[2 marks]**

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0 2 . 5 Explain why intensive agriculture may be very productive but not very efficient. **[2 marks]**

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10

0 3 A group of students wants to investigate the effect of vegetation cover on rain-splash soil erosion.

0 3 . 1 Describe how they could carry out the investigation in the laboratory. **[5 marks]**

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03 . 2

Give **three** variables, other than those you have standardised, that affect rain-splash erosion in the field.

[3 marks]

1

2

3

03 . 3

Explain why it is easier to obtain valid results in the laboratory than in the field, and give **one** disadvantage of laboratory studies.

[2 marks]

.....

10

04 . 1

Explain **four** properties of water that make it essential for life.

[4 marks]

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0 4 . 2

Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

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0 4 . 3

Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

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10

0 5

Table 1 shows the number of plant species per m² in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m ²	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

0 5 . 1

Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

0 5 . 2

Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

0 5 . 3

Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

0 5 . 4

Explain how burning maintains heathland.

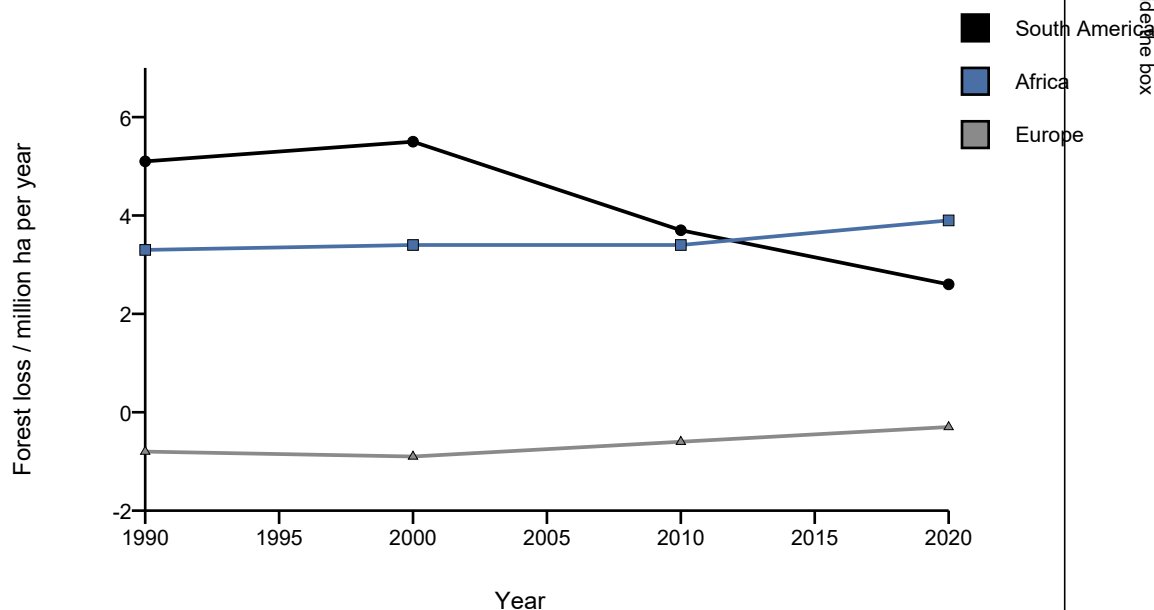
[2 marks]

10

0 6

Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 6

1 Describe the trends shown in Figure 1.

[3 marks]

0 6

2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

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06 . 3

Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

06 . 4

Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

10

07 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1

2

07 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

07 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

5

08 . 1

Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

08 . 2

Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

1

2

3

08 . 3

Explain how auditory monitoring can be used to study bats.

[2 marks]

08 . 4

Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

10

09 . 1

Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

09 . 2

Explain how soil organisms provide services that humans depend on.

[3 marks]

09 . 3

Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

09 . 4

Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

10

10 . 1

Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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Extra space

10 . **2** State what is meant by **industrial symbiosis**.

[1 mark]

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10

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Write an essay on one of the following topics.

1	1	.	1
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Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

1 1 . 2

Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

END OF QUESTIONS